

SOC L1 Log Investigation Practical Test

Includes realistic SIEM-style alerts, questions, answers, and explanations to prepare students for SOC analyst roles.

Scenario 1 — Authentication Log Analysis

SIEM Alert: Multiple Failed Login Attempts (Event ID 4625) from IP 185.221.10.45 targeting user 'administrator' with 320 attempts in 4 minutes.

1. What type of attack is likely occurring?

Answer: Brute Force Attack

Explanation: A large number of login attempts against a single account in a short time is a classic brute force attempt.

2. Which log field indicates the attacker's source?

Answer: Source Network Address / Source IP

Explanation: This field identifies where the login attempt originated from.

3. What does Event ID 4625 represent?

Answer: Failed Logon Attempt

Explanation: Windows Security Event ID 4625 indicates that a login attempt failed due to invalid credentials.

4. What should the SOC analyst check next?

Answer: Check if any successful login (Event ID 4624) occurred after the failures.

Explanation: Attackers often eventually succeed if passwords are weak.

5. What mitigation could prevent this attack?

Answer: Account lockout policy and Multi-Factor Authentication.

Explanation: Lockout policies stop repeated attempts and MFA prevents unauthorized access even if password is guessed.

Scenario 2 — Suspicious Login Alert

User john.doe logged in from India and then Russia within 2 minutes via O365.

1. What security indicator does this represent?

Answer: Impossible Travel

Explanation: It is impossible for a user to log in from two distant locations within such a short time.

2. What attack might have happened before this login?

Answer: Phishing or Credential Stuffing

Explanation: Attackers likely obtained the user's credentials through phishing or leaked passwords.

3. What should the SOC analyst check immediately?

Answer: Login history, IP reputation, and MFA status.

Explanation: These help confirm if the login is legitimate or malicious.

4. What immediate response action should be taken?

Answer: Force password reset and revoke active sessions.

Explanation: This prevents attackers from maintaining access.

Scenario 3 — Email Security Alert

Phishing email detected from security-update@microsoft-support.com with attachment Salary_Adjustment_2025.xlsm.

1. What type of attack is this?

Answer: Phishing Attack

Explanation: The email attempts to trick the user into opening a malicious attachment.

2. Identify two indicators showing the email is malicious.

Answer: Spoofed domain (microsoft) and SPF/DKIM/DMARC failures.

Explanation: Attackers often slightly modify domains and authentication failures indicate spoofing.

3. What risk does the .xlsm attachment indicate?

Answer: Malicious macro execution.

Explanation: XLSM files can run macros that execute malware when opened.

4. What should SOC do if the user opened the attachment?

Answer: Isolate the endpoint and scan for malware.

Explanation: This prevents potential malware from spreading.

Scenario 4 — Endpoint Security Alert

EDR detected mimikatz.exe accessing lsass.exe.

1. What attack technique is being used?

Answer: Credential Dumping

Explanation: Mimikatz extracts credentials stored in system memory.

2. What sensitive data is the attacker trying to extract?

Answer: Passwords, NTLM hashes, and Kerberos tickets.

Explanation: These allow attackers to authenticate as other users.

3. Which Windows process stores authentication credentials?

Answer: lsass.exe

Explanation: LSASS manages authentication and stores credential material.

4. What attack stages may follow this activity?

Answer: Privilege Escalation and Lateral Movement.

Explanation: Stolen credentials allow attackers to move across the network.

Scenario 5 — Internal Network Alert

RDP connection from HR-Laptop-14 to FileServer-02 using Administrator account at 03:12 AM.

1. What attack phase does this indicate?

Answer: Lateral Movement

Explanation: The attacker is moving from one compromised host to another system.

2. Why is the login suspicious?

Answer: It occurred at an unusual time using an admin account from a user laptop.

Explanation: Admin logins from non-admin devices are suspicious.

3. What previous attack likely enabled this?

Answer: Credential Dumping or Phishing.

Explanation: Attackers likely obtained admin credentials earlier.

4. What logs should SOC investigate next?

Answer: Windows security logs, authentication logs, and RDP logs.

Explanation: These help determine the full attack path.

Scenario 6 — Network Traffic Alert

Firewall log shows 4.8 GB HTTPS data transfer from Finance-PC-03 to external IP at 02:41 AM.

1. What suspicious activity could this represent?

Answer: Data Exfiltration

Explanation: Large outbound data transfers may indicate stolen data leaving the network.

2. Why is the time important?

Answer: Activity occurred outside normal business hours.

Explanation: Attackers often operate at night to avoid detection.

3. What should SOC investigate next?

Answer: User activity, file access logs, and destination IP reputation.

Explanation: This helps confirm whether data theft occurred.

4. What category of incident might this become?

Answer: Data Breach Incident.

Explanation: Unauthorized transfer of sensitive information could lead to a breach.

Final SOC Analyst Decision Scenario

Alert chain: Phishing email → User opened attachment → Mimikatz executed → Admin login → RDP to file server.

1. Identify the attack stages in order.

Answer: Initial Access → Credential Dumping → Privilege Escalation → Lateral Movement.

Explanation: The attacker progresses through multiple stages of the kill chain.

2. What should SOC L1 escalate to SOC L2?

Answer: Credential compromise and lateral movement activity.

Explanation: These indicate a serious security incident.

3. What type of major attack could this lead to?

Answer: Ransomware attack or major data breach.

Explanation: Attackers often deploy ransomware after gaining administrative control.